

Functional requirements

FEATURE 1 → Create Post

Input

```
{  
  "authorId": 1,  
  "content": "Hello"  
}
```

Flow

Validate user exists

Save post

Return response

FEATURE 2 → Add Comment

Flow

Human Comment

Save comment

Update virality +50

Bot Comment

Before saving:

Check Horizontal Cap

Check Cooldown Cap

Check Depth Limit

After saving:

Increment bot counter

Virality +1

Trigger notifications

FEATURE 3 → Like Post

Flow

Save like

Virality +20

Non-Functional Requirements

A. Concurrency Safety

Problem:

200 bots comment simultaneously.

Without atomic operations:

Bot 1 reads count = 99

Bot 2 reads count = 99

Both insert

Final = 101

FAIL

Solution

Redis atomic commands:

INCR

SETNX

EXPIRE

EXISTS

These operations are thread-safe.

B. Statelessness

The app server should NOT store:

HashMap

static counters

in-memory lists

Because:

multiple server instances

horizontal scaling

distributed systems

Everything temporary goes into Redis.

C. Scalability

Why Redis?

Because:

millions of operations

low latency

distributed locking

atomic counters

D. Atomicity

Meaning:

Either:

all validations pass

OR nothing happens

No partial updates.